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Superseding
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MILITARY SPECIFICATION

CONTROL, INTEGRATED ELECTRONIC CENTRAL

C-4850/ASQ-85

This specification has been approved by the
Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall provide control of the UHF communications, UHF auxiliary receiver, the UHF Data Link Communications, Digital Data Communications Set and the UHF ADF equipment specified in 6.7.

1.2 Associated Equipment - This equipment shall operate with the associated equipment listed in 6.7.

2. APPLICABLE DOCUMENTS

2.1 General - The documents listed in 6.8 of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

*3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. Parts and materials requirements shall conform to Specification MIL-E-5400.

*3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

FSC 5895

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3.3 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.3.1 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 11 pounds.

3.3.2 Reliability

3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 1000 hours without the necessity for readjustment of controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have total operating life of 3000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.3 Cabling and Connections -

3.3.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

3.3.3.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.3.4 Control Panels - The control panel shall conform to the applicable requirements of Specification MIL-C-6781. The configuration of the control panel must be approved by the procuring activity prior to preproduction testing.

*3.3.5 Interchangeability - The equipment shall meet the interchangeability requirements of Specification MIL-E-5400.

3.3.6 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

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3.3.7 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.3.8 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.3.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25°C ±5°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	5.0 VAC ±2.0% and 27.5 ±0.5 VDC

3.3.10 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 1A equipment, except as modified herein:

3.3.10.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curve I of MIL-T-5422E (Figure 2) as limited by Curve III (5G maximum acceleration).

3.3.11 Warm-Up Time - There shall be no warm-up time required.

3.3.12 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amounts.

(1) AC Power (Single Phase) 5.0V, Category "B" 14 VA
(2) D.C. Power, 28V, Category "B", 0.1 Amp.

3.3.12.1 Degraded Performance - Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electric system operation. Operation shall return to normal with no resulting damage to the equipment.

3.4 Performance

3.4.1 The equipment shall be of the Console type in accordance with Specification MIL-C-6781, unless otherwise specified herein, and shall furnish the following requirements.

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*3.4.1.1 Function - The equipment shall provide control of the following equipments to the extent so specified.

(1) UHF Communications (AN/ARC-51A) - It shall be possible to select manually any of the 3,500 frequencies available on this equipment and to preset up to 20 channels on any of these available frequencies. Additionally, it shall be possible to select a fixed tuned guard channel on this equipment, normally 243.00 MCS.

(2) UHF Data Communications (AN/ARC-124) - It shall be possible to select manually any of the 3,500 frequencies available on this equipment and to preset up to 19 channels on any of these available frequencies.

(3) UHF Auxiliary Receiver (R-1286/ARR-69) - It shall be possible to select any of the 20 UHF auxiliary receiver channels between 265 and 284.9 Mc/s. In addition, it shall be possible to select one fixed tuned guard channel on this equipment, normally 243 Mc/s.

(4) Digital Data Communications (AN/ASW-27) - It shall be possible to control the following functions relevant to the operation of the AN/ASW-27.

a. Address Select
b. Cancel Reply, Forced Reply, and Normal Reply
c. Anti-jam
d. Primary Power

*3.4.1.1.1 Provisions shall also be available for the following:

(1) Dual Operation of the UHF Communications and the UHF Auxiliary Receiver.
(2) Select operation of the UHF ADF equipment when so desired.
(3) Select either of two UHF antennas for UHF Data Link and either of two UHF antennas for UHF Communications
(4) Supply switching logic to enable two ID-758/ARC Frequency/Channel Indicators to display the number of the preset channel selected on the UHF Communications (AN/ARC-51A).

3.4.1.2 Form Factor - The unit shall be console mounted in accordance with Specification MIL-C-6781. Overall size, excluding connectors, shall not be greater than 5-3/4 inches in width, 8-5/8 inches in length and 6-1/2 inches in depth. Controls shall not extend beyond the panel face more than 1-1/8 inches.

3.4.1.3 Panel Front - An edge lighted front panel in accordance with the applicable requirements of MIL-P-7788 shall be provided. The layout of the controls shall be as shown in Figure 1.

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3.4.1.4 Controls - The following controls shall be located and suitably marked on the front panel.

*3.4.1.4.1 UHF Comm MAIN and AUX.

(1) Three thumbwheels (MAN FREQ) for manually selecting any one of the 3,500 available frequencies on the UHF communications (MAIN).

(2) A twenty position rotary switch (MAIN CHANNEL) for selecting any one of twenty predetermined frequencies for the UHF communications (MAIN).

(3) A three position switch for selecting GUARD, PRESET, or MANUAL operation of the UHF communications (MAIN).

(4) A dual concentric control (VOLUME-SQUELCH DISABLED) for disabling the squelch circuits within the UHF communications (MAIN) and for adjusting the amplitude of the audio signals delivered to the output circuit by the UHF communications (MAIN).

(5) A five position rotary switch (MAIN-AUXILIARY) for selecting the mode of transmission and reception of the AN/ARC-51A and auxiliary receiver AN/ARR-69 as follows:

AN/ARC-51A	AN/ARR-69
OFF	OFF
T/R	ADF
TR+G	ADF
ADF	CMD
ADF	G

(6) A twenty position thumbwheel switch (AUX CHANNEL) for selecting any one of the twenty UHF auxiliary receiver channels.

(7) A dual concentric control (AUX VOL-OFF/SENS) for controlling the application of the audio output of the UHF auxiliary receiver for external use, (OFF) and the adjusting the amplitude (VOL) of the audio signals delivered to the output circuit by the UHF auxiliary receiver. In addition, it shall be capable of controlling the sensitivity (SENS) of the auxiliary receiver.

(8) A three position toggle switch shall permit automatic or manual selection of either of the two UHF antennas for voice reception and transmission.

*3.4.1.4.2 UHF Data Link Communications (AN/ARC-124)

(1) Three thumbwheels (MAN FREQ) for manually selecting any one of the 3,500 available frequencies on the UHF Data Link Communications.

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(2) A twenty position rotary switch (DATA LINK CHANNEL) for selecting any of 19 predetermined frequencies on the UHF Data Link Communications, or a manual position on which any of the 3,500 available frequencies can be selected.

*3.4.1.4.3 Digital Data Communication AN/ASW-27

(1) A dual Concentric eight position rotary switch for selection of 64 octal coded addresses for the Data Communications Systems AN/ASW-27.

(2) A three position lever-locked toggle switch labeled "Reply" for controlling Data Link Reply status. The center position shall be labeled "NORM" and the two lever-locked positions shall be labeled "CANCEL" and "FORCED".

(3) A two position toggle switch labeled "ANTI-JAM" with the "ON" position lever-locked. Abbreviations for the positions of the switch shall be "ON" and "OFF". In addition to the signal required, the "ON" position of this switch shall provide for power on switching for the auxiliary equipment simultaneously with the selection of "ON".

(4) A two position (ON OFF) switch for power to the AN/ARC-124, AN/ASW-27, and Data Link Auxiliary Equipment.

(5) A two position toggle switch to permit selection of either of two UHF antennae for data link reception and transmission.

*3.4.1.5 Controls, Internal - Provisions shall be made for changing the preset frequencies on the respective memory drums of the UHF Communications (AN/ARC-51A) and the UHF Data Link Communications AN/ARC-124 by removing the unit from normal installation. Doors on the dust cover of the unit will provide access to the frequency drums for presetting frequencies.

*3.4.1.6 Antenna and Unattenuated Audio Output Switching - Provisions shall be made to provide switching of the attenuated audio output of the AN/ARC-51A and AN/ARR-69 for antenna selection when the UHF communications is in T/R or T/R+G. When the mode switch (3.4.1.4.1(5)) is in the T/R - ADF or T/R+G ADF position the communications antenna shall be connected to the AN/ARC-51A and the ADF antenna to the AN/ARR-69. The unattenuated output of the AN/ARR-69 shall be switched to the same pin of the output connector when the UHF communication is in ADF and the Auxiliary Receiver is in CMD or Guard. This function shall be incorporated in the UHF Communication main and AUX control (3.4.1.4.1).

3.4.1.7 Voice Output Switching - Provisions shall be made to provide switching of the attenuated audio outputs of the AN/ARC-51A and AN/ARR-69 for use by the intercom system. The audio output of the AN/ARC-51A shall be applied to the output circuit when the UHF comm is in

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T/R or T/R+G. The audio output of the AN/ARR-69 shall be applied to the output circuit when the UHF comm is in ADF and the Auxiliary Receiver is in CMD or Guard. This function shall be incorporated in the UHF Communication - MAIN AUX CONTROL (See3.4.1.4).

3.4.1.8 Auxiliary Receiver Sensitivity Switching - Provisions shall be made to provide grounding of the Auxiliary Sensitivity control output when the Auxiliary Receiver is in ADF. This function shall be incorporated in the UHF Communication - MAIN - AUX CONTROL (See 3.4.1.4).

3.4.1.9 Electrical Connections - Connections to external circuits shall be provided as follows as approved equivalent:

Reference Designation	Receptable Type	Function
J1	MS 3112 E-22-55 PW	General
J2	MS 3112 E-22-55 P	General

*3.4.1.10 Torque Requirements

(1)The torque requirements to activate the switches of 3.4.1.4.1(2), 3.4.1.4.1(5) and 3.4.1.4.2(2) shall be between 1.0 and 6.0 inch pounds.

(2) The torque requirements to activate the switch of 3.4.1.4.1(1), 3.4.1.4.1(6) and 3.4.1.4.2(1) shall be between 1.0 and 3.0 inch pounds.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction (First Article) Tests
(2) Initial Production Tests
(3) Acceptance Tests
(4) Life Tests

4.2 Preproduction (First Article) Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the responsibility of the contractor and shall be

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conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction (First Article) Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

4.2.3 Preproduction (First Article) Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of noncompliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first 10 production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and

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quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the Initial Production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

*4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests
(2) Sampling Tests
(3) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product
(2) Operational Test
(3) Manufacturing Run In Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship require-

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ments have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run In Test - Each equipment shall be operated under the conditions specified herein for a period of 10 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ±3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 10-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 10-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 10-hour period specified above may be composed of two 5-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the government inspector in accordance with the following:

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	0*
Next 50	1
Next 75	1
Next 100	1

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	1 each additional 200 or fraction thereof

*NOTE: If by contract action the Initial Production Test (4.3) is deleted, then a Sample Test shall be conducted on one equipment from the first 10 produced.

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

(3) Manufacturing run in test specified in paragraph 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.3.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity).

4.4.3.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

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4.4.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occuring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following. (Equipments which have successfully passed the Initial Production Test, Sampling Tests, or Special Tests may be selected for life tests.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal ground (0-5000 ft.)
Humidity	Room ambient
A.C. Voltage	5.0 V ± 2.0%
D. C. Voltage	27.0 ± 2.0 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the

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equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number
(2) The circuit reference symbol number
(3) The part function
(4) Name of the manufacturer
(5) Nature of the failure
(6) The number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

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4.8 Presubmission Testing - No item, part of complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

5. PREPARATION FOR DELIVERY

*5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specifications MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in Specification MIL-E-17555, the method of preservation for level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

6. NOTES

6.1 Intended Use - The Control, Integrated Electronic Central C-4850/ASQ-85 covered by this specification is intended to be used in Naval aircraft to control the equipments identified in paragraph 6.7.

6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 +0.5V DC and 5.0 + 2.0%V AC as applicable. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity or design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.4 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

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(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

*6.5 Type Designations - The type designation may be modified by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.8. The correct type number shall be used on nameplates, shipping records and instruction books, as applicable.

*6.6 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number and date of this specification.

(2) Selection of applicable levels of packaging and packing (see 5.1).

*6.7 Associated Equipment - The equipment shall operate with the following associated equipment which shall not be supplied as part of the equipment:

Radio Set AN/ARC-51A

Radio Set AN/ARC-124

Receiver, Radio AN/ARR-69

AN/ARA-50

Communications Set, Digital Data AN/ASW-27

Frequency/Channel Indicator ID-758/ARC (2 each)

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*6.8 Applicable Documents -

SPECIFICATIONS

Military

MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-C-6781	Control Panel: Aircraft Equipment, Rack or Console Mounted
MIL-P-7788	Plate, Plastic, Lighting
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-C-22670	Control, Radio Set C-2790/ARC; Indicator, Frequency Channel ID-758/ARC; Mounting MT-2141/ARC and Switching Unit SA-629/ARC

STANDARDS

Military

MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of

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*6.9 Availability of Documents -

(1) When requested specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Publications and Forms Center, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania.

6.10 Revisions - In specification revisions and superseding amendments an asterisk "" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

Project Number 5895-N113

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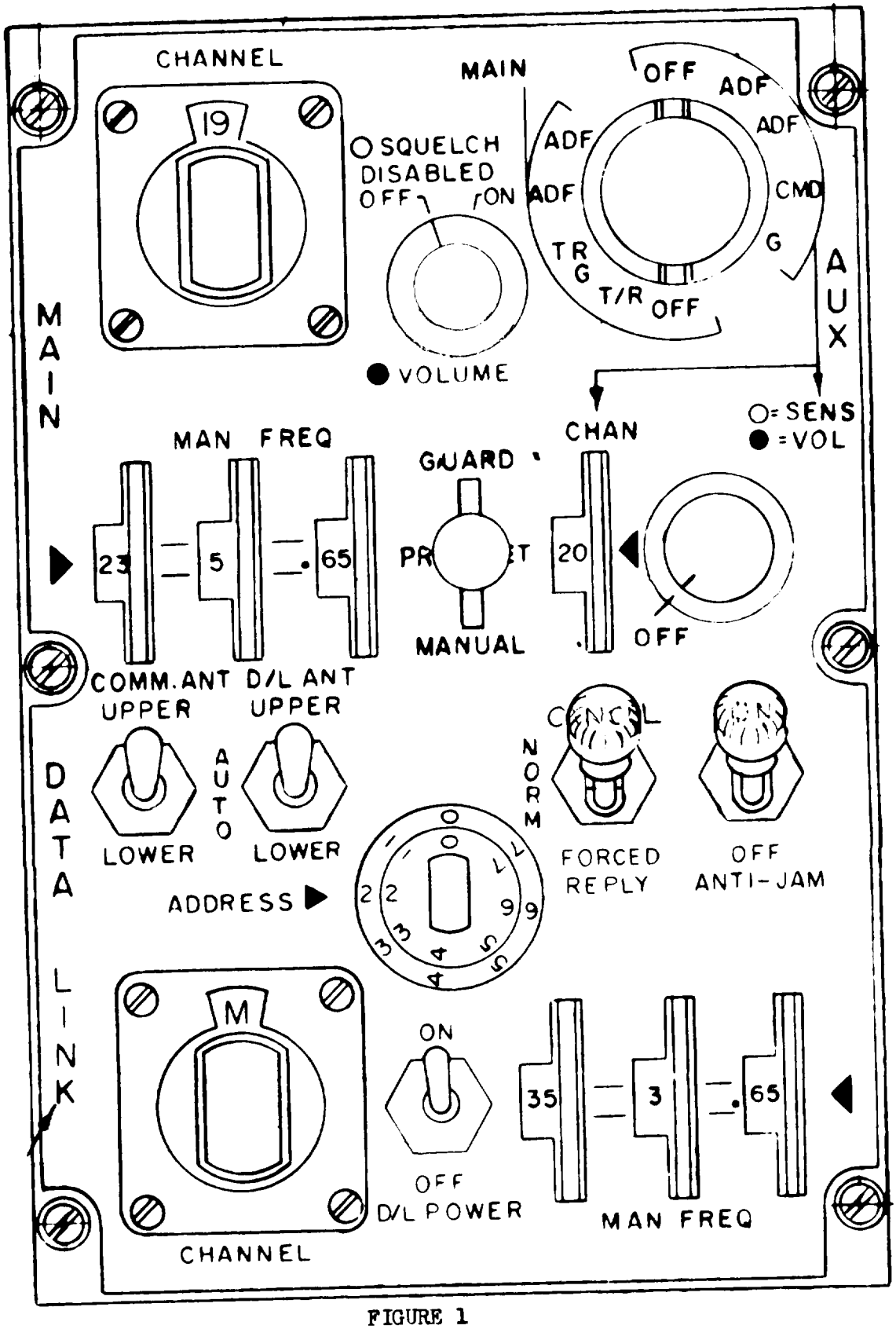


FIGURE 1

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-C-23768A(AS) CONTROL, INTEGRATED ELECTRONIC CENTRAL C-4850/ASQ-85

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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